



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 1 • STANDARD 5.MD.C.5

# Math is Explaining and Sharing

Lesson 1-4 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can construct an argument using equations, drawings, or words, and use volume to defend a recommendation.

**Language Objective:** I can explain my thinking with clear math vocabulary using the words argument, conjecture, and cubic units.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

argument

conjecture

counterexample

volume

cubic unit



Tap any word to see what it means and a picture.

1

— A chain of reasons — equations, drawings, or words — used to defend your thinking.

2

— A mathematical statement you believe is true but have not yet proven.

3

— A single example that shows a statement is not always true.

4

— The amount of space a solid figure takes up, measured in cubic units.

5

— A cube with edges 1 unit long — the unit used to measure volume.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How can the group determine the amount of soil needed, and how can they check that their answer is accurate?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **argument** — A chain of reasons — equations, drawings, or words — used to defend your thinking.

*argumento — Una cadena de razones — ecuaciones, dibujos o palabras — que usas para defender tu pensamiento.*

- ☐ **conjecture** — A mathematical statement you believe is true but have not yet proven.

*conjetura — Una afirmación matemática que crees verdadera pero que aún no has probado.*

- ☐ **counterexample** — A single example that shows a statement is not always true.

*contraejemplo — Un solo ejemplo que muestra que una afirmación no siempre es verdadera.*

- ☐ **volume** — The amount of space a solid figure takes up, measured in cubic units.

*volumen — La cantidad de espacio que ocupa una figura sólida, medida en unidades cúbicas.*

- ☐ **cubic unit** — A cube with edges 1 unit long — the unit used to measure volume.

*unidad cúbica — Un cubo cuyas aristas miden 1 unidad — la unidad que se usa para medir el volumen.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**To find how much soil is needed, find the \_\_\_\_ of the garden beds. Because volume covers three dimensions, label the answer in \_\_\_\_ feet. Finding the volume two different ways and getting matching totals shows the solution is \_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** Three large boxes hold  $240 \times 3 = 720$  cubic inches, and two jumbo boxes hold  $400 \times 2 = 800$  cubic inches, so  $800 > 720$ .

**Evidence:** A strong answer calculates both totals (720 and 800) and compares them to recommend the jumbo boxes. A weak answer picks an option without computing and comparing both volumes.

**Reasoning:** This evidence connects the definition of argument to the mathematical claim.

#### Start

##### Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- To find how much soil is needed, find the \_\_\_\_ of the garden beds. Because volume covers three dimensions, label the answer in \_\_\_\_ feet. Finding the volume two different ways and getting matching totals shows the solution is \_\_\_\_.
- My answer is \_\_\_\_.

*Mi respuesta es \_\_\_\_.*

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#### Build

##### Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.  
*Mi afirmación es \_\_\_\_.*
- I know because \_\_\_\_.  
*Lo sé porque \_\_\_\_.*
- So \_\_\_\_.  
*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** argument
2. **Notes 2:** conjecture
3. **Notes 3:** counterexample
4. **Notes 4:** volume
5. **Notes 5:** cubic unit
6. **Watch for this mistake:** *Reporting volume without cubic units — saying the garden needs '48 feet' of soil, which is a length, or '48 square feet', which is an area.*
7. **Try It 1:**
8. **Try It 2:**